



Attentional differences in driving judgments for country and city scenes: Semantic congruency in inattention blindness

Kristen Pammer*, Caroline Blink

The Australian National University Canberra 0200, Australia

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ABSTRACT

'Looked-but-failed-to-see' vehicle collisions occur when a driver gives all indications of having responsibly evaluated the driving situation yet still fails to see a hazard that is clearly in view. The experience maps well onto the psychological phenomenon called inattention blindness (IB). IB occurs when a viewer fails to see an unexpected object that is clearly visible, particularly if they are concentrating on an additional primary task. In this study, a driving-related IB task was used to explore whether an unexpected stimulus (US) such as a pedestrian or animal, is more likely to be seen in country or city-related driving scenarios if it is congruent or incongruent with the semantic context of the scenes, and thus congruent or incongruent with the attentional set of the viewer. Overall, participants were more likely to see the US in the City scenarios, which also demonstrated a borderline effect of congruency, with incongruent stimuli less likely to be seen than congruent stimuli. Analyses suggested that driver experience was related to detection of the US in City scenarios but not Country scenarios. However, analyses also revealed that participants generally tended to drive in city rather than country environments, thus prompting speculation that the results may reflect attentional requirements for familiar and unfamiliar driving scenarios. Thus we suggest that the analysis of the driving situation, and the attentional set that we develop to filter information, change when the driving situation is more familiar.

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1. Introduction

The driving situation is one which requires constant monitoring of the visual environment, and continuous filtering of visual information in order to attend to the most important cues. In many cases however, we fail to see something of critical importance while driving. Referred to as 'looked-but-failed-to-see' accidents (Hills, 1980; Treat, 1980), they have been implicated as the third most frequent type of driver error (Brown, 2005). Such accidents have been attributed to a number of different factors, for example insufficient visual search strategies, where the driver fails to systematically move their eyes around the visual environment to capture relevant visual information (Summala et al., 1996), or attentional load (Hancock et al., 1990; White and Caird, 2010). Another explanation for looked-but-failed-to-see-accidents is that the 'unseen' item does not fit the driver's current attentional expectations. Such a situation could occur when a cyclist comes from an unexpected direction (Summala et al., 1996), or when a highly visible vehicle such as a police car, is sitting in an unexpected location (Langham et al., 2000).

Cognitively, the looked-but-failed-to-see experience appears to map onto a known psychological phenomenon called inattention blindness (IB). In IB, an observer fails to see an unexpected stimuli or event while attending to another, primary task (Mack and Rock, 1998). In one of the earliest versions of IB, Neisser and Becklen (1975) participants were required to follow one of two superimposed videos. In one video, participants were required to follow an informal basketball play, and ignore a 'hand-slapping' game similar to the child's 'patty-cake' game. In the other participants followed the hand-slapping game, and ignored the basketball game. In each of the unattended situations, an unusual event occurred, such as swapping the gender of one of the basketball players, or having the 'slapping hands', stop, shake, and resume play. Very few of these unusual events were detected by the participants, suggesting that while attending to one situation, unrelated events in the same visual space go unnoticed by the viewer.

In one common version of IB, participants are presented with a primary attention task in the form of a briefly presented, masked cross (Mack and Rock, 1998). The cross appears approximately five times, and in each trial, one arm of the cross is longer than the others. At each trial the participants are required to indicate which arm of the cross was longer. On about the fifth trial, an additional unexpected stimulus (US) appears such as a small square, and IB is determined by whether the participant notices the US. This is

* Corresponding author.

E-mail address: Kristen.pammer@anu.edu.au (K. Pammer).

the static version of IB. In the dynamic version, the primary task consists of objects such as circles and triangles moving randomly around the computer screen. In this case, the participants' task is to count how many times the tracked items bounce off the edges of the computer screen, and the US is an additional object moving a set course through the centre of the display. In general, approximately 25% of people fail to notice the additional US in these tasks, however this number varies depending on the nature and location of the US. For example, the US is noticed far more frequently if it is salient to the participant such as their name or a 'smiley face' (Mack and Rock, 1998), but failure to notice it increases dramatically when the primary task is made more difficult (e.g., Cartwright-Finch and Lavie, 2007).

The general finding – that attending to a primary task decreases a viewer's tendency to detect an additional unexpected stimulus – is a consistent result, and many physical parameters of the phenomenon have been explored, such as the nature of the primary task (Beanland et al., 2011; Koivisto and Revonsuo, 2008; Most et al., 2000, 2001), the colour/shape/location and size of the US (e.g., Mack and Rock, 1998), the role of eye movements (Beanland and Pammer, 2010a), experience (Beanland and Pammer, 2010b), and the difficulty of the primary task (e.g., Cartwright-Finch and Lavie, 2007; Lavie, 1995).

Outside the IB literature, it is also known that the 'attentional set' of an observer is important for the detection of a target in experimental settings. Here, attentional set refers to the recognition that we generate a mental template for objects, situations or behaviours that have consistent semantic features and feature relationships. Attentional set can be simple, such as looking for a red circle amongst red and blue shape distractors, where the attentional set would be "red" + "circle". A complex attentional set is more akin to Bartlett's concept of schema (e.g., Bartlett, 1932) where the attentional set for driving for example, would involve the objects and procedures involved in the driving situation. Folk et al. (1992) demonstrated that participants detecting a target take advantage of a cue that is consistent with a specific attentional set, for example when participants know that a target is defined by a particular colour, a cue defined by that colour is more effective than a cue defined by some other feature. Such expectation is similarly important in the detection of the US in IB. Most et al. (2001, 2005) employed a dynamic IB task in which the primary task was to monitor black and white objects – such as circles and squares – moving randomly around the computer screen. IB was determined by whether the participant noticed an additional US, such as a "+" sign traverse the display through the moving circles and squares. In one experiment, participants were required to track the black items and ignore the white items or the reverse – track the white items, ignoring the black. Rates of IB were much higher when the US was congruent with the colour of the items to be tracked, i.e., if participants monitored the black items, they were far more likely to see the US when it was also black, than if the US was white. The suggestion here was that participants were more likely to see the US when it shared a common feature with the tracked items, because the attentional set in this case was "black" and black items gained priority access to higher order recognition systems.

Most et al. (2005) further demonstrated an IB effect for faces, in order to determine if the same attentional set mechanism was applicable for complex features, as well as simple features such as shape and colour. In a dynamic IB display consisting of four Caucasian faces and four African American faces moving randomly around the computer screen, participants were required to track either the Caucasian or African American faces. Participants were more likely to see an US when it was consistent with the race of the face that they had been asked to monitor.

Koivisto and Revonsuo (2008) through another series of dynamic IB experiments systematically manipulated the features

of the distractor and attended items while keeping the US constant. Their question was whether participants established an attentional set for the features of the attended items and ignored the unattended items, or conversely, coded the unattended items in order to selectively ignore them and concentrate on the attended items. They concluded that participants developed an attentional set consistent with the features of the attended items, not the unattended items.

Overall, these results suggest that observers are able to establish an attentional set that makes salient the features of the objects that are explicitly attended. In this then, an additional unexpected item in a visual scene is more likely to be seen when it shares one or more features of the attended items. However in general this research has been in the context of setting an attentional set for stimulus features such as colour and/or shape. One question that arises as a consequence of this research is whether realistic settings engender a similar attentional set. For example, given a city scene, are we more likely to see an US if it is congruent with the attentional set – such as a man in a business suit, compared to an US that is incongruent with a city schema, such as a deer or tractor?

Set theory is the theoretical framework for the concept of attentional set. Such a model derives from a number of different studies that identify the cognitive advantage incurred by semantic relationships between objects. For example, in early research, Biederman et al. (1982) demonstrated automatic activation of the semantic relationships between objects that appeared to occur in parallel to identification. Such results illustrate the power of contextual information in object identification. Set theory, would therefore predict that a 'city' attentional set would include items such as buses, buildings, cars, people, but would be unlikely to include tractors, sheep or hay silos, which would be more likely to be included in a 'country' attentional set. Thus we would predict that if set theory works for semantic schemas in the same way as for features, then IB would be higher when the US is a sheep in a city scene, compared to when the US is a man in a business suit in a city scene because the sheep does not fit our attentional set for the city.

It is well known that we can extract the salient features of a scene extremely quickly and accurately; Potter (1975), presented participants with 16 scenes presented as an RSVP (rapid serial visual presentation) task. Here participants were able to correctly identify a target scene even at the fastest presentation rate of 125 ms if they knew beforehand what would be in the scene (e.g., a child). This rate of performance was consistent with subjects who were previously shown the actual target scene. Thus participants were equally accurate in detecting a scene if they knew exactly what they were looking for, compared to whether they simply knew what would be in the scene. This result shows the speed with which our attentional system can extract relevant information from a scene at a glance.

There is currently very little research looking at IB and the attentional set of the observer. Koivisto and Revonsuo (2007) presented participants with an array of pictures in which they were required to identify either the animals, or the furniture in the array. The US in this case was the presentation of printed word that was either congruent or incongruent with the semantic category that they were attending to. Their results demonstrated that detection for the US was higher when the attended items and the US were congruent, supporting the suggestion that detection of an unexpected stimulus is enhanced not only when its features match the attended items, as shown previously, but also when it fits an attended semantic category.

We argue that there is a conceptual link between IB and looked-but-failed-to-see-accidents, such that IB could provide the theoretical framework for the experience of such accidents. However, again, there is little research exploring incidents of IB in the driving situation. The only study that we know of is by Most and

Astur (2007). In this study participants were required to drive through a virtual environment, following indicator signs designated either blue or yellow. Participants followed the yellow signs, ignoring the blue, or followed the blue signs, ignoring the yellow. The assumption was that the participant would develop an attentional set for blue or yellow, depending on which sign they were following. The US was a motorcyclist that was either blue or yellow such that it was either congruent or incongruent with the attended signs. Drivers were slower to brake for the motorcycle when it was the incongruent colour, and were also more likely to 'collide' with it. Thus the results remain consistent with the idea that we are more likely to see an unexpected stimulus when it fits a predetermined attentional set. Although this study combines the notion of attentional set with IB and driving, the link between the attended stimuli and the US is again featural – in this case colour. It still remains to be seen whether a particular attentional set as defined by a real world scenario engenders the same response. Thus, if when driving we develop an attentional set for the environment in which we are driving, such as the city or country, then set theory would predict that this attentional set would mediate our detection of hazards when driving. Moreover, positioning the study within the context of IB allows us to extend the IB literature to include the possibility that set theory can extend from simple features to real-life contexts. In this sense then, the study is designed to address a point of theoretical significance in the cognition of IB, as well as the practical element of hazard detection when driving.

What then do we know within the driving literature about the degree to which different driving environments engender different attentional requirements? Again, the literature is limited. Borowsky et al. (2012) demonstrated that eye movements and hazard detection was different depending on whether participants were driving in urban compared to residential environments. Although these researchers were interested in driver experience, they nevertheless demonstrated that pedestrian-related hazards were detected more frequently in residential environments, and eye fixations for the curb and the road varied depending on whether it was the urban environment or the residential environment. Similarly, Borowsky et al. (2008) presented experienced and inexperienced drivers with typical road signs (e.g., No Right Turn) either on the expected right side of the road or the unexpected left side of the road. Experienced drivers – who presumably have a well-developed attentional set for driving, were significantly better at detecting road signs in the expected location compared to the unexpected location. Moreover, the inexperienced drivers showed no difference in detecting signs presented on the 'correct' right side, or 'incorrect' left side of the road. This is consistent with the inexperienced drivers having a less well-defined attentional set for the driving environment compared to experienced drivers. These findings suggest that it is possible to develop a driving attentional set that dictates different driving requirements. But again, we remain unenlightened as to whether a particular attentional set would facilitate or hinder detection of an unexpected stimulus. In the current study we present participants with a static IB task in which 'city' or 'country' driving scenes are varied between subjects, and the US is either congruent or incongruent with the scenes. Consistent with set theory we would predict that incongruent stimuli – in this case the presence of a kangaroo in the city – would be more likely to produce IB (less likely to be seen) as it does not fit the attentional set generated by the city scene. From this we could also conclude that set theory extends beyond the development of an attentional set for simple features, to the development of an attentional set for semantic information in the form of semantic schemas. Moreover, in the current study participants will make consecutive judgments about the relative safety of a driving scenario, allowing us to tentatively explore whether a tendency towards caution when driving induces its own attentional set. One possibility is

that participants who are more cautious and thus more likely to describe driving situations as "unsafe", may have a heightened sense of vigilance, and more likely to detect the US. Finally, driving experience was recorded. It is well known in the driving literature that increases in driving experience change visuo-attentional mechanisms such as increasing search speed, eye movement and fixations (Underwood et al., 2011a,b; refer to Konstantopoulos et al. (2012) for a review), as well as increased hazard detection (e.g., Underwood et al., 2005). Thus it is not unreasonable to predict that detection of the US is mediated by driving experience. The current findings are an attempt to understand the generation and flexibility of attentional-set when driving in specific environments, with the practical significance in terms of the context in which drivers are vulnerable to lapses of attention.

2. Method

2.1. Participants and design

Ninety-two undergraduates participated voluntarily, either for course credit or \$20AUS. Participants were excluded if they had participated in an IB study before (3 participants) or failed to see the US on the full attention trial (one participant). The remaining 88 participants (43% female; $M_{age} = 22$ years, $SD = 4.5$) were randomly assigned to the four experimental conditions. This experiment used a 2 (environment: city vs. country) $\times$ 2 (unexpected stimulus: kangaroo, businessman) between-subjects design. All participants provided informed consent and had normal or corrected-to-normal vision.

2.2. Apparatus and stimuli

Visual stimulus were displayed on a large screen 100 cm $\times$ 50 cm LG monitor controlled by a Dell laptop, with a photograph of a driving situation taking up 80% of the screen. Participants sat approximately one metre from the screen.

The experiment consisted of ten 2-s trials, with the first three practice trials and five control trials presented in random order. The practice and control trials contained photos of driving scenes from either the country or the city, which were constructed using photos taken from the driver's perspective, with elements added and pedestrians removed using Adobe Photoshop Elements. The critical trial (trial 9) and full attention trial (trial 11) contained an extra item, the US (see Fig. 1). The US was either a brown kangaroo (approximately $2.9^\circ \times 4.2^\circ$ VA) or brown-coated businessman (approximately $4.5^\circ \times 1.7^\circ$ VA) located on the right side of the photo, subtending approximately 12.9° VA from the centre point of the scene, and was present throughout the 2-s the photo was displayed. Effort was made to equate the featural components of all city and country trials, while maintaining the basic visual integrity of a city or country driving situation. Similarly, pilot studies were conducted to find the best visual match in terms of size, colour, etc., for the businessman US and the kangaroo US. Each trial was preceded by a grey screen with a fixation cross in the centre and followed by a mask consisting of randomly placed black lines.

A note about the choice of stimuli is important here. Canberra – where the study was conducted – has a relatively small population spread over a small geographical area. While the city centre is typical with malls, plazas, and office blocks, a short drive from the city centre will reach the bush countryside, and suburban Canberra is criss-crossed with wildlife corridors. Moreover, kangaroos are a common occurrence in suburban and country roads. The ACT government reports more than 2000 kangaroo-related car accidents around Canberra each year, with 17% of Canberrans having hit a kangaroo with their car. Thus the stimulus selection in the form of

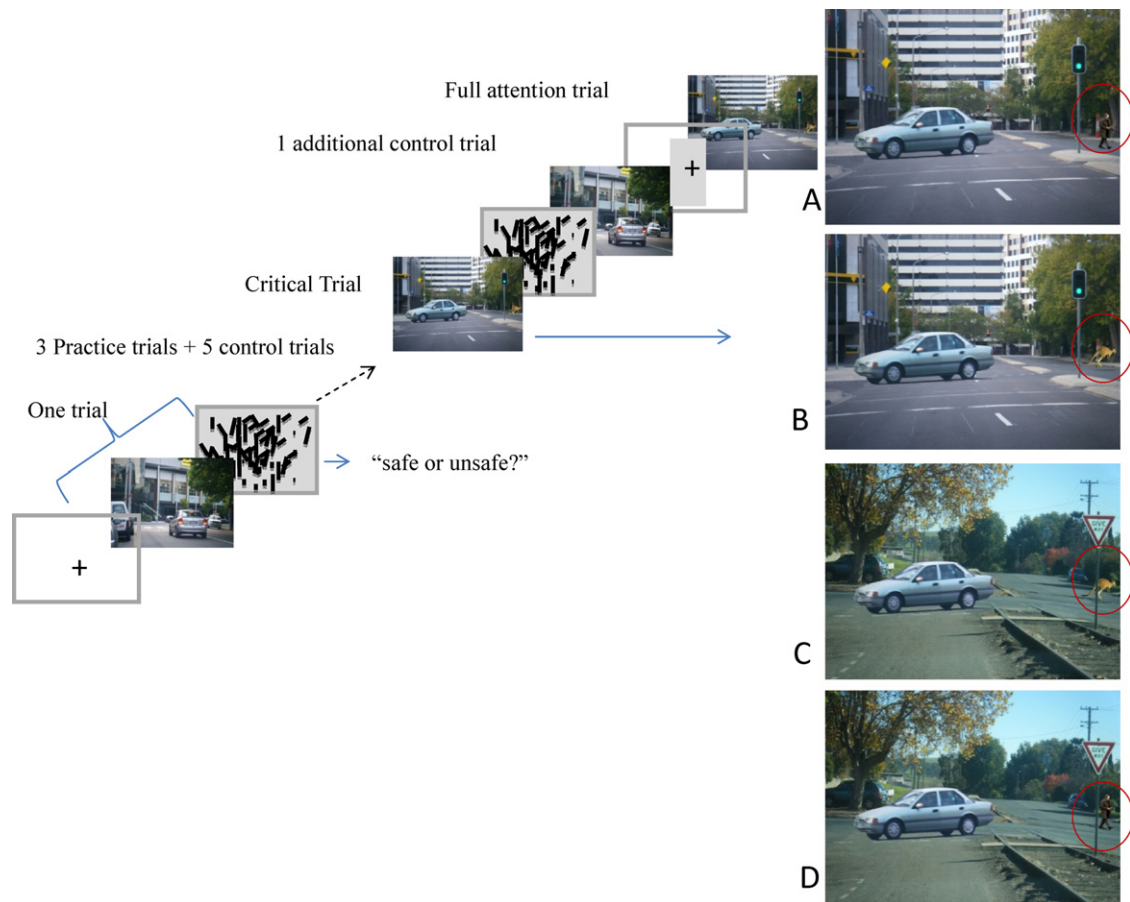


Fig. 1. The experimental sequence consisted of a fixation followed by a driving decision image flashed for 2 s. After the mask the participants were required to explain whether the image they had just seen was a safe or unsafe driving situation. Images to the right are the four different critical trials, where A = city scene and the congruent US, B = city scene and the incongruent US, C = the country scene with the congruent US, and D = the country scene with the incongruent US.

a business man and a kangaroo to be congruent with the city and country respectively, are not unusual or trivial, but in fact reflect the expected driving experiences of the participants. It is difficult to quantify how incongruous each of the stimuli is in each environment. In this case we made a judgement call; neither stimuli are unusual per se, but a man in a business suit is much more likely in the city centre than in a country town. Similarly while kangaroos are common in Canberra, they are rare (but not unknown) in the city centre.

2.3. Procedure

Participants were tested individually and were told that we were interested in their ability to detect hazards in different environments, specifically city versus country environments and that the first task involved looking at a series of photos depicting a number of driving scenes from the country or the city. In each condition participants saw only city or country scenes for all trials.

Participants were positioned approximately one metre from the television screen (consistent with the average distance from the driver to the windscreen when driving) and asked to imagine that the widescreen television screen was the windscreen of their car and they were looking out at the scene displayed. After each photo, a decision was to be made indicating whether the photo showed a safe or unsafe driving situation, both for them as the approaching vehicle as well as for other drivers on the road. They pressed the 1 key to indicate the scene was safe and 2 to indicate unsafe after which they were asked to briefly explain their decision to the experimenter before moving on to the next photo. This was the

primary task and was chosen as it was consistent with decisions that drivers make in a typical driving situation i.e., whether the current driving situation safe to proceed. The task was not intended to be a proxy for the actual response to a safe or unsafe driving situation, but to engage the cognitive mechanisms involved in making such decisions. The technique is consistent with other studies, such as Underwood et al. (2011a,b) and standard hazard perception tests which require the participant indicating on a touch screen the presence of a hazard (e.g., Anstey et al., 2012).

Following the critical trial, participants were asked if they had noticed anything other than the cars, street signs, trees or houses in the country environment conditions (or car, trees, buildings or traffic lights in the city environment conditions). Participants who noticed the US were asked to describe what they saw and indicate where in the photo they had seen the item. They were then shown a forced-choice sheet containing six shapes and asked to identify which of these they had seen in the photo. The shapes (cyclist, businessman, kangaroo, horse, farmer, and shopper) were chosen for their congruity with either a country or city environment. The non-noticers (those participants who failed to see the US) were informed that on the previous trial an additional item had been present in the photo and that it was one of the six shown on the forced-choice sheet and if they had to guess which one would they say it was. We have adopted the use of a force choice sheet in all our IB studies to determine, first of all if there may be any implicit processing of the US. This would be reflected in non-noticers having an increased tendency to choose the US from a list of likely targets. Secondly, it allows us to determine whether there is any bias towards choosing one stimuli over another. An analyses of

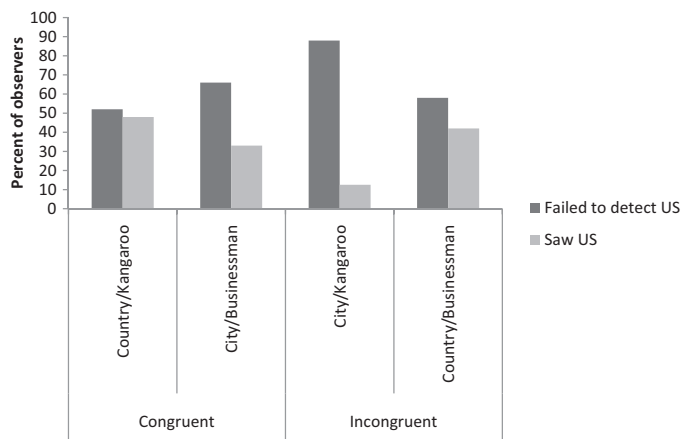


Fig. 2. Rates of IB for the four road tasks background environment/unexpected stimulus.

the choices made by participants who did not report seeing the US indicated a significantly greater tendency towards selecting the cyclist in all conditions, followed by the business man and then the shopper $\chi^2(3, N=64)=40.5, p<.001$. Another standard control trial was then run before the full attention trial. For the full attention trial, participants were told that they did not need to judge whether the driving situation in the next photo was safe or unsafe, all they needed to do was to look at the photo. The full attention trial was designed to determine that the US is clearly detectable when the participant is simply looking at the scene and not attending to the primary task. Participants were then asked to complete a brief questionnaire about their driving experiences which included demographic questions.

3. Results

Unlike most IB tasks, accuracy judgments for the primary task were not analysed as there was no explicit correct or incorrect response for each trial. Participants were required to make a “safe” or “unsafe” judgment on each trial with an explanation for their decision that was recorded by the experimenter. A typical response would be; “safe, because the car had stopped at a red light”. The aim was to simply get participants to attend to the primary task. All participants made valid and logical decisions on all trials.

The descriptive statistics for participants between conditions, for relevant factors, are presented in Table 1.

3.1. IB for congruent and incongruent judgments

The overall rates of IB are presented in Fig. 2. Pearson Chi-square indicated borderline significance for the Road-Task $\times$ IB interaction $\chi^2(3, N=95)=7.56, p=.056$.

Inspection of Fig. 2 suggests that overall rates of IB were higher in the city than in the country, but there is only a borderline effect of congruency. This marginal effect of congruency appears to be carried by the City condition, where there were higher rates of IB in the incongruent city condition (City/Kangaroo) compared to the congruent city condition (City/Businessman). This was confirmed statistically, in that the interaction between detection of the US and congruency condition was borderline significant in the City condition, $\chi^2(1, N=48)=3.0, p=.07$ and not significant in the Country condition.

However, inspection of Fig. 2 suggests that the largest effect here is between City and Country scenes, in that it would appear that there is a higher rate of IB in the city compared to the country

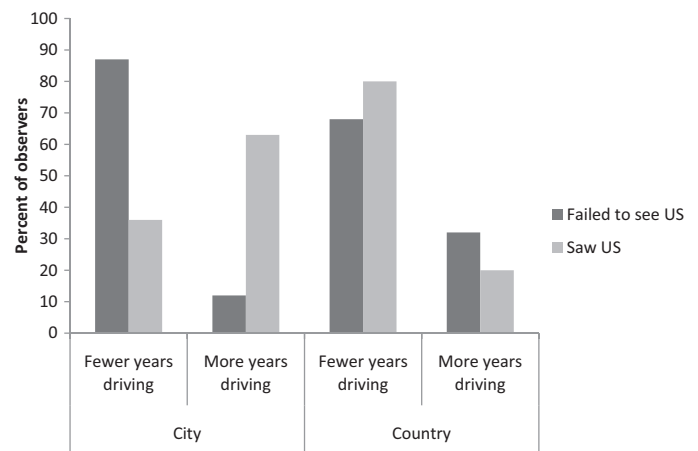


Fig. 3. Rates of noticing the US categorized by driving experience for the City and Country conditions.

in general. This was confirmed statistically $\chi^2(1, N=95)=5.036, p<.05$, where the results were collapsed over congruency.

The results thus far would suggest that congruency is only of marginal significance, and only in the city scenes. What appears to be more important is whether the participant was exposed to the city or the country driving scenes where it would appear that there are higher rates of IB in the city compared to the country. The subsequent results are therefore divided into these City or Country driving categories.

In order to explore why there might be a difference between City and Country scenes, analyses were conducted evaluating the influence of years of driving experience, and whether the type of judgments participants made could have primed participants towards detecting the US.

3.2. Driving experience

Logistic regression was performed to determine whether years of driving experience reliably predicted noticing the US. However, because the above analyses suggest that the differences between participants lie with whether they experienced the City or Country scenes, analyses were conducted separately for these two conditions.

In both the City and Country analyses, the predictor variable was driving experience in years, for noticing the US on the critical trial. In the City scenes the model was able to correctly classify 96% of observers who failed to see the US, and 36% of observers who saw the US for an overall success rate of 81.4%. The model indicates that driving experience significantly predicts IB in the city scenes, refer to Table 2. The Chi-square analyses indicate that participants with less driving experience were significantly less likely to see the US than those with more driving experience $\chi^2(1, N=43)=11.2, p<.002$, refer to Fig. 3. There was a difference in the average experience for noticers ($N=11$, mean = 7.3 years driving, $SD=4.7$) and non-noticers ($N=32$, mean = 3.8 years driving, $SD=3.6$), but this was of borderline significance $t(41)=-2.3, p=.04$, and also reflected the fact that three times as many people failed to notice the US.

In the Country scenes, the full model with the predictor was not reliably different from a constant-only model, indicating that in country scenes years driving did not predict whether a participant saw the US, refer to Fig. 3. There was also no difference between driving experience for noticers ($N=19$, mean = 2.7 years driving, $SD=1.6$) and non-noticers ($N=20$, mean = 3.5 years driving, $SD=5.4$)

Table 1
Descriptive statistics for the four conditions.

Factor	Condition				p value
	Country/Kangaroo (Congruent)	Country/Businessman (incongruent)	City/Businessman (Congruent)	City/Kangaroo (incongruent)	
Gender, % female	57	60	54	59	.97
Age, mean (SD)	21.9 (4.8)	21.4 (5.02)	22 (3.9)	23 (4.6)	.65
Driving experience in years, mean (SD)	2.85 (1.6)	3.4 (5.5)	4 (4.2)	4.8 (3.9)	.43
Typical driving environment, % City, % Country, % Both	5, 85, 10	0, 75, 25	0, 78, 17	0, 72, 20	.54

Table 2
Logistic regression of US noticing for city scenes and country scenes.

Predictor	Condition	B	S.E. β	Wald χ^2	df	Odds ratio	p
Driving experience	City	.197	.092	4.6	1	1.217	.032
	Country	-.072	.098	.54	1	.93	.462

Overall the results indicate that the more driving experience a participant had, the less likely they were to miss the US (i.e., experience IB), however this effect was only significant for the City.

3.3. Type of judgments typically made

Each of the decision control trials were designed to be somewhat ambiguous in terms of their safety so that the participant was required to attend to the picture and make a coherent judgment. Overall participants made 10 'safety' judgments: Three practice trials, five control trials, the control trial following the critical trial and the full attention trial. Thus each participant had a number of trials out of 10 that they rated as either safe or unsafe. The variable TotalUnsafe was added to a logistic regression predicting noticing the US. For the City environment, the variable was not significant, whereas it was in the Country environment, where the model was able to correctly classify 85% of observers who failed to see the US, and 38% of observers who saw the US for an overall success rate of 63.8%, refer to Table 3. The Chi-square analyses indicated that participants who were more likely to judge a scenario as safe, were significantly less likely to see the US than those participants who made mainly unsafe judgments $\chi^2(1, N=47)=9.26, p=.002$, refer to Fig. 4.

As is clear from Fig. 4, participants in the city condition were far more likely to make "safe" decisions than "unsafe" decisions, however this did not vary with whether or not they were likely to see the US. In the Country condition however, people who were more likely to make "safe" decisions were more likely to miss the

US, whereas those people who were more likely to make "unsafe" judgments, were more likely to see the US.

4. Discussion

When monitoring a visual stimulus, a viewer is more likely to notice an unexpected object appearing if it shares some features with the stimulus being monitored. It is suggested that when attending to a visual stimulus we develop an attentional set for the object of our attention (Most et al., 2005), and objects that share features of this attentional set are more likely to be seen (Folk et al., 1992). The aim of the current study was to look at IB in a realistic situation such as driving, manipulating the semantic congruency of an unexpected item with the driving scenario. It was suggested that if participants are more likely to detect (have lower incidences of IB) a target that was congruent with the semantic context, then this would be consistent with set theory, and would extend previous findings in IB to suggest that an attentional set can be established for semantic context as well as stimulus features. The results indicated a borderline effect for congruency that was apparent in the City scenes but not in the Country scenes, such that participants were more likely to see the US in the City scene when it was a man in a business suit (detection rate was 33%) compared to a kangaroo (detection rate was 12%). However as this was only of borderline significance and occurred only in the City driving condition, the result should be treated with caution. In the Country condition, there was no effect of congruency, participants were equally likely to detect (or conversely, fail to detect) the US whether it was the kangaroo which was congruent with the scene (48% detection rate), or the man in the business suit which was incongruent with the scene (42% detection rate). This result suggests that there may have been inherent attentional differences between city and country scenarios, and this was confirmed by the fact that there was a significantly higher rate of IB in city scenes compared to country scenes irrespective of the nature of the US. This finding is likely to be mitigated by the fact that the majority of participants reported that they engaged in mostly city driving (77%) compared to mainly country driving (3%) or both equally (18%). So it may be possible to reconceptualise this study as not specifically 'City' and 'Country' driving, but 'familiar' and 'unfamiliar' driving environments. This point will be considered in more detail later in the discussion. One way or another, there was clearly a difference between the City and Country conditions. In order to explore this further, years of driving experience and/or the type of decisions that a participant was making overall was considered. The findings suggested that both these factors were influential, but in different ways for the City and Country conditions. Experienced drivers were more likely to see the US in the City condition, but driving experience made no difference

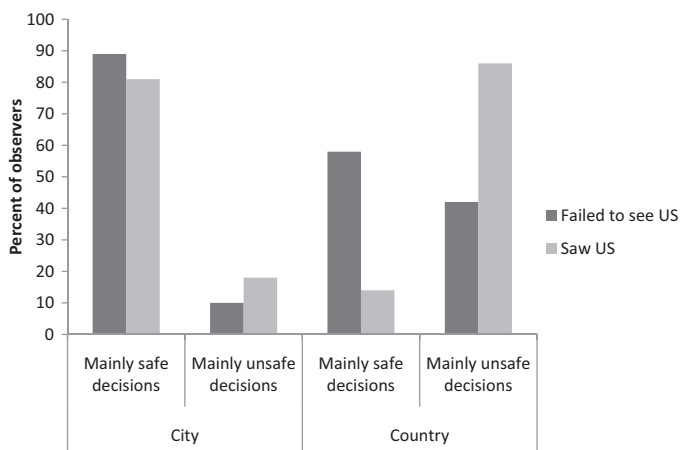


Fig. 4. Incidences of noticing the US based on whether participants were more or less likely to judge driving environments as safe or unsafe.

Table 3

Logistic regression of US noticing for city scenes and country scenes with the predictor as the number of total unsafe judgments the participant responded with.

Predictor	Condition	β	S.E. β	Wald χ^2	df	Odds ratio	p
Total unsafe judgments	City	.142	.228	.388	1	1.15	.533
	Country	.725	.275	6.97	1	2.065	.008

to rates of noticing in the Country condition. Similarly, in the City condition there was a tendency to make “safe” rather than “unsafe” judgments compared to the Country, but this did not contribute to whether a participant was more or less likely to detect the US. Conversely in the Country condition, participants’ judgments did contribute to their detection of the US, such that people who were more likely to respond to trials as being “unsafe”, were also more likely to detect the US.

We must entertain the possibility that the differences observed between City and Country’s driving scenarios are in fact due to familiarity, such that the City and Country conditions were in fact proxies for familiar and unfamiliar driving environments. In this case, the latter result demonstrating that participants in the Country condition are more likely to rate the driving environment as ‘unsafe’ and then also more likely to detect an US, raises the intriguing possibility that drivers in an unfamiliar environment – such as the country for predominately city drivers – are more likely to rate an environment as unsafe which in turn increases their vigilance making them more likely to detect an US. This is good news for when drivers move out of their driving comfort-zone, but bad news for drivers when they are within their familiar driving areas. [Charlton and Starkey \(2011\)](#) investigated familiarity when driving. More accurately they were interested in the development of automaticity when driving, so here we make an assumption that familiarity is more-or-less synonymous with their concept of automatic processing. We make this assumption because in their study they operationalise the development of automaticity in their participants by having participants ‘drive’ in a simulator a section of road repeatedly over three months. Therefore in their scenario automaticity is achieved by increasing familiarity with the driving situation. After 20 sessions in a simulated driving scenario for the same piece of road, they demonstrated that participants were increasingly less likely to detect changes in the environment (such as road markings or the deletion of specific landmarks), but were more likely to detect a specific object (a particular car that changed colour and location from trial to trial). Interestingly, regarding the former finding, detection for changes in the environment increased (spiked) when the simulation added an unfamiliar piece of road to the road they had become used to driving on. They also reported that drivers found the scenarios less difficult as they became more familiar and were more likely to ‘drift-off’ when driving. Similarly, [Borowsky et al. \(2008\)](#), demonstrated that inexperienced drivers – who would notionally be less familiar with the driving situation – were less impaired in their detection of road signs presented on the ‘incorrect’, left side of the road, compared to experienced drivers who were far more likely to miss signs presented on the unfamiliar left side. [Martens and Fox \(2007\)](#) also measured the eye movements of drivers over five simulator sessions. They found that participants were less likely to glance at road signs as the scenario became more familiar over sessions, and although they were better at detecting signs than control drivers, they were less likely to detect when a sign had been changed.

Persevering with the notion that ‘city’ and ‘country’ may in this instance be proxies for ‘familiar’ and ‘unfamiliar’ environments, it was also demonstrated in the current study, that driving experience was related to noticing the US, with more experienced drivers more likely to notice the stimuli in the ‘familiar’ city condition compared to the ‘unfamiliar’ country condition. There is a large literature looking at driving experience, (refer for example to [Borowsky](#)

[et al., 2012, 2010, 2009, 2008](#) or [Chapman and Underwood, 1998; Underwood, 2007](#)). However, we were unable to find any specific research looking at the relationship between familiarity of the driving situation and driver experience. Nevertheless, in general, experienced drivers are better at detecting and responding to hazards than less experienced drivers ([Borowsky et al., 2009; McKenna and Crick, 1991; Summala, 1987](#)), have more efficient scanning strategies when driving ([Chapman and Underwood, 1998](#)), are better at estimating hazards or risky situations ([Brown and Copeman, 1975](#)), are less likely to engage in risky driving behaviour ([Gusfield, 1991](#)), and are more likely to engage in a wholistic analysis of the driving situation ([Benda and Hoyos, 1983](#)). Combined with the literature of familiarity discussed above, we can surmise that experience was related to noticing in the familiar City scenarios because drivers may be able to draw on two semantic templates – the *driving experience* per se, as well as the City environment. However, in an unfamiliar Country environment, where experience was unrelated to detecting the US, we would have expected the experienced drivers to have been more likely to detect the US for all the reasons described above – they would be more likely to scan the environment more effectively and made better assessments of risk. Clearly this is an area that requires further research.

If then, the research conditions become reconceptualized in terms of familiarity, what does this mean for the original research question – whether set theory as a mechanisms for attentional selection, extends beyond physical feature selection, to the semantic features of an attentional set? The non-significant result for the more unfamiliar Country condition, may now make more sense, as by definition, an unfamiliar environment will have fewer specific exemplars defining it as a consistent schema. Again this is consistent with research looking at object identification in natural scenes and the contribution of contextual information. [Biederman et al. \(1982\)](#), showed that information that is schema consistent with a target object, such as the probability of finding it in the scene, and its size, are accessed in a single glance (simulated by a 125 ms presentation), and are likely to be processed in parallel with the actual identification of the object (i.e., the processing of its unique features). In the IB literature, familiarity appears to add little to the detection of the US ([Mack and Rock, 1998](#)). [Devue et al. \(2009\)](#), found no difference in the detection of the US when it was a familiar face, compared to an unfamiliar face, or the participant’s own face. However the difference between these studies and the current study is in the description of exactly what is familiar. In the IB studies of [Mack and Rock \(1998\)](#), and [Devue et al. \(2009\)](#), the specific familiarity of the US was manipulated and found not to capture attention. Here however, it is the familiarity of the stimulus scene itself that may be important.

In the current study we set out to determine the some of the factors that may be important in detecting unexpected items or events when driving in a city or country environment. While we recognize that stylized snapshots of city and country driving scenes in no way capture the complexity of the real driving situation, we nevertheless provided tentative support for the idea that we develop an attentional set when driving that is consistent with a semantic schema of the driving situation. However in analysing the driving profiles of the participants, we recognise that our ‘city’ and ‘country’ scenes may be proxies for familiar and unfamiliar driving environments, in that the majority of the participants reported driving mainly in city environments. This then suggests that in the

current project, the more familiar city driving scenarios engendered higher levels of IB, but were dependent on driving experience where more experienced drivers were more likely to detect the target. There was also a higher tendency to report driving situations as safe, and a greater susceptibility to the congruency effect, where a congruent businessman was more likely to be seen than an incongruent kangaroo. Conversely, drivers in the more unfamiliar country driving scenarios were more likely to detect an unexpected object, irrespective of whether it was congruent or incongruent with the environment, and this occurred for both novice and experienced drivers, and they were more likely to rate driving scenarios as unsafe which was related to an increased tendency to detect the US.

How do the current findings relate to road safety? It is likely that the development of a specific attentional set is an unavoidable and in fact necessary consequence of driving. The important practical outcomes in this case are to foster an increased awareness that our familiarity with the driving situation strengthens our attentional set, and this has consequences for our ability to detect hazards and our vigilance. However the current finding also has implications for other road users. For example, motorcycle accidents are over-represented in road accident statistics, thus we are currently exploring the possibility that motorcycles are less likely to fit out attentional set when driving thus translating into a decreased tendency to detect a motorcycle compared to a car

The results from the current study suggest that the familiarity of the driving situation may be important in the way in which we tune attentional mechanisms such that we become more vigilant in an unfamiliar environment, but because we lack an attentional template for items that should exist in the environment, this vigilance does not manifest behaviourally into an increased tendency to spot unexpected items on the road. Conversely, in a familiar environment we may be less vigilant, but counter intuitively, more likely to detect an unexpected item particularly when it matches the semantic context. These conclusions would benefit greatly from subsequent research, for example using eye movement measurements to support our suggestion of increased vigilance in the different environments. Similarly in subsequent studies we intend to move beyond IB to hazard detection, where we can manipulate the number of hazards and the semantic congruency of the hazards to the environment. Finally, the notion of familiarity in this study was developed post hoc, on inspection of the data collected. To develop this further, 'familiarity' for the driver should be established systematically and possibly individually tailored to allow a clear and methodical manipulation of the concept. This is also linked with the fact that we acknowledge that the choice of stimuli here are limited. We argue that they are appropriate for the question and the participant sample we used, and do not obviate the conclusions we make in regards to the theory. However we recognize the specific US examples are unlikely to be generalisable to other populations. A more extensive study would include creating stimuli sets with a greater range of semantic congruency, and ideally tailored to the drivers most familiar/unfamiliar driving situation. Thus looked-but-failed-to-see accidents appear to be dependent on the driving situation, with particular emphasis on the familiarity of the driving environment and the experience of the driver.

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